CAMBRIDGE
INTERNATIONAL EXAMINATIONS

**June 2003**

**INTERNATIONAL GCSE**

MARKING SCHEME

MAXIMUM MARK: 80

**SYLLABUS/COMPONENT: 0620/02**

**CHEMISTRY**

**(Core Paper 2)**

root 19

**1 (a) (i)** Fe/Cu ALLOW Zn [1]
**(ii)** C/N/S/F/C*l*/Br [1]
**(iii)** O/S [1]
**(iv)** C [1]
**(v)** Li/Na/K ALLOW F [1]
**(vi)** CU/Zn/Br/Kr [1]

**(b)** argon - light bulbs;
chlorine - kills bacteria;
carbon - as lubricant;
helium - in balloons [4]

**(c) (i)** covalent [1]
**(ii)** $BrF_5$ ALLOW $F_5Br$ [1]
**(iii)** ions/charged particles;
NOT: particles
not free to move in solid/free to move in molten/liquid state [2]

**2 (a)** drop small tube in acid/loosen string/idea of mixing zinc and acid/let go of cotton
ALLOW: cut the string [1]
NOT: heat (the acid)
NOT: pull the string

**(b) (i)** correct plotting including 0-0 point (– 1 per omission or error) [2]
**(ii)** best curve drawn and to go through origin [1]
**(iii)** no more gas produced/reaction finished;
all zinc reacted/used up [2]

**(c)** graph drawn with faster initial rate and starting at 0-0;
ALLOW: straight line as initial rate
ends up at 55 $cm^3$ [2]

**(d) (i)** 2 (HC*l*) [1]
**(ii)** zinc chloride [1]
**(iii)** 136 [1]
IGNORE units

**(e)** substance containing only one type of atom/substance which cannot be broken down to any other substance by chemical means [1]
NOT 'can't be split' alone
NOT is a pure substance

**3 (a) (i)** evaporation/vaporisation/boiling [1]
**(ii)** freezing/solidification [1]
NOT: fusion
**(iii)** condensing/condensation/liquefaction [1]

**(b)** 2nd box ticked [1]

**(c)** A;
energy needed to overcome forces between molecules/idea of energy input/ taking in heat [2]

**(d) (i)** chlorine [1]
**(ii)** bromine [1]
**(iii)** sodium chloride [1]


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**(e)** **(i)** diffusion [1]
NOT: Brownian motion

**(ii)** ammonium chloride [1]
NOT: ammonia chloride

**(iii)** ammonia diffuses or moves faster/HC*l* diffuses or moves slower/ammonia has lower mass/HC*l* higher mass/molecules of HC*l* and ammonia move at different speeds [1]
NOT: ammonia evaporates faster/HC*l* evaporates more slowly

**(f)** neutralisation/acid base [1]
NOT: exothermic
NOT: addition

**(g)** **(i)** thermometer [1]

**(ii)** reference to the solid or melting point of the solid is needed for the mark. boiling point of water too low to get solid to melt/boiling water cannot get to 155°C [1]
NOT: boiling point of water is only 100°C/boiling point of water too low.
NOT: water boils off first

**(iii)** so that the liquid is the same temperature throughout/no hot or cold spots/so the tube is the same temperature as the thermometer/so heat can circulate in all places [1]
ALLOW: so that temperature of liquid is balanced
NOT: to keep temperature constant

**4** **(a)** **(i)** <u>breaking down</u> of molecules substances using <u>heat</u> [1]

**(ii)** substance which speeds up a reaction [1]
NOT: alters/changes rate of reaction
NOT: speeds up and slows down rate

**(b)** ethene/ethylene [1]
NOT: formula

**(c)** **(i)** paraffin [1]

**(ii)** 4000g/4kg [1]
(correct unit needed)

**(iii)** $C_2H_4$; $H_2$ [2]

**(d)** **(i)** two units polymerised with continuation bonds at either end and hydrogen atoms drawn [1]
ALLOW: $-CH_2CH_2CH_2CH_2-$
ALLOW: $-[-CH_2CH_2-]-_n$
ALLOW: $-[-CH_2-]-_n$

**(ii)** addition (polymerisation) [1]

**5** **(a)** (sodium) hydroxide/ammonia; → green/grey green; [2]
silver nitrate; → yellow; [2]
ALLOW: lead nitrate NOT: cream
ALLOW: bubble chlorine → grey/black (precipitate)
silver nitrate; → white: [2]
barium chloride/nitrate; → white; [2]
ALLOW: lead acetate


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**(b)** filtration/filtering or diagram of correct apparatus for filtration (filter paper must be present on diagram)
NOT: decanting
sodium chloride through filter paper/shown on diagram;
NOT: filtrate through filter paper
evaporate off water from sodium chloride/suitable diagram [3]
ALLOW: distilling off water

**(c)** different atoms/elements
(chemically) joined/bonded/combined (both points needed)
(reference to mixtures = 0 unless qualified enough in time frame e.g. a mixture of elements which are then chemically combined) [1]

**(d) (i)** chlorine/$Cl_2$ [1]
**(ii)** sodium/Na [1]

**6 (a)** potassium/magnesium/aluminium [1]

**(b)** they did not have electricity/did not know about electrolysis/did not know the metal existed [1]
NOT: did not have the right technology

**(c) (i)** indication that bubbles produced rapidly or quickly/slower than magnesium but faster than zinc
OR number of bubbles produced intermediate between magnesium and zinc; [1]
uranium dissolved slower than magnesium but faster than zinc/dissolves at medium rate etc. [1]
**(ii)** atoms of same element with different mass number/different number of neutrons/different nucleon number [1]
NOT: compounds/molecules with different mass number
**(iii)** indication of use for energy – nuclear power stations/nuclear energy [1]
ALLOW: atomic/nuclear bombs
NOT: curing cancer/medical uses
NOT: 'for fuel'

**(d)** magnesium oxide
ALLOW: MgO [1]

**(e) (i)** idea of mixture of (different) metals [1]
**(ii)** alloys harder/stronger/decreased malleability/increased toughness/increased corrosion resistance/heat or electrical resistance increased [1]
NOT: increase in melting point
NOT: cheaper
NOT: improving properties

**(f)** removes oxygen from zinc oxide [1]
ALLOW: definition of reduction involving oxidation numbers/electron transfer

**(g) (i)** reversible reaction [1]
ALLOW: equilibrium
**(ii)** 76-80% [1]

**(h) (i)** correct electronic structure of Mg (2.8.2) on diagram [1]
**(ii)** loses two electrons/loses its valence electrons = 2
forms $Mg^{2+}$ ion = 1
loses electron(s) = 1 [2]
forms $Mg^{2+}$ ion by losing electrons = 2


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